

PIVOT TABLES



In this section we'll cover exploring & analyzing data with **pivot tables**, Excel's essential "drag & drop" tool that lets you perform calculations without formulas or functions

TOPICS WE'LL COVER:

Data Structure

Creating Pivot Tables

Calculation Options

Sorting & Filtering

Calculated Fields

GOALS FOR THIS SECTION:

- Understand the source data requirements for analyzing data with pivot tables
- Quickly explore different pivot table views
- Summarize and show values in different ways
- Sort and filter pivot table results
- Create basic calculated fields

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IMPORTANT: DATA STRUCTURE

Data Structure

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Pivot Tables

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For pivot tables to work properly, your source data must have a **tabular structure**:



TABULAR

Opportunity	Sales Agent	Region	Product	Deal Stage	Deal Won	Close Date	Close Value
PE84CX4O	Zane Levy	West	GTX Basic	Closed	1	3/2/2017	517
9ME3374G	Vicki Laflamme	West	MG Special	Closed	1	3/2/2017	57
M6WEIXCQ	Rosalina Dieter	West	MG Advanced	Closed	1	3/6/2017	3284
902REDPA	Cassey Cress	East	GTX Pro	Closed	0	3/9/2017	0
5J9CMGDV	Elease Gluck	West	MG Special	Prospecting			
JJXR8R6	James Ascencio	West	GTX Plus Pro	Closed	0	3/17/2017	0
CSK2JP1H	Violet Mclelland	East	GTX Plus Basic	Closed	1	3/11/2017	1014
SBCR987L	Kami Bicknell	West	GTX Basic	Closed	1	3/23/2017	590
UP409DSB	Anna Snelling	West	MG Advanced	Engaging			
AO9Z2D17	Violet Mclelland	East	GTX Plus Pro	Closed	0	3/11/2017	0
5M58DTJK	Elease Gluck	West	MG Special	Closed	1	3/5/2017	58
KNY1OSAB	Anna Snelling	West	GTX Pro	Closed	1	3/14/2017	4899
EAZDUUM9	Moses Frase	East	MG Advanced	Closed	0	3/1/2017	0
JYKM0B00	James Ascencio	West	GTX Pro	Closed	1	3/6/2017	4338
KU28360J	Kary Hendrixson	West	GTX Basic	Closed	1	3/19/2017	578
N4SD17JR	Reed Clapper	East	GTX Basic	Closed	1	3/1/2017	556

- Single table (*fields as columns, records as rows*)
- Clear column headers
- Each column has a single data type
- No calculated fields or subtotals



FREE FORM

	EAST							
	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Opportunities	226	181	172	206	207	134	212	190
Deals won	130	145	84	123	160	67	112	133
Win rate	58%	80%	49%	60%	77%	50%	53%	70%
Revenue	\$353K	\$447K	\$208K	\$302K	\$409K	\$192K	\$269K	\$364K

	WEST							
	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Opportunities	263	212	202	254	201	181	270	224
Deals won	138	170	85	147	164	89	152	187
Win rate	52%	80%	42%	58%	82%	49%	56%	83%
Revenue	\$337K	\$482K	\$195K	\$375K	\$440K	\$245K	\$374K	\$370K

- Multiple tables (*months as columns, metrics as rows*)
- Multilayered column headers
- Each column contains different data types
- Calculated fields (*win rate*)

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WHY PIVOT TABLES?

Pivot tables are one of the most powerful tools for data analysis, as they allow you to easily filter, summarize, and analyze information without modifying the raw data

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5M58DTJK	Elease Gluck	West	MG Special	Closed	1	3/5/2017	58
KNY1OSAB	Anna Snelling	West	GTX Pro	Closed	1	3/14/2017	4899
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N4SD17JR	Reed Clapper	East	GTX Basic	Closed	1	3/1/2017	556



Region East

	Month							
Values	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Opportunities	226	181	172	206	207	134	212	190
Deals won	130	145	84	123	160	67	112	133
Win rate	58%	80%	49%	60%	77%	50%	53%	70%
Revenue	\$353K	\$447K	\$208K	\$302K	\$409K	\$192K	\$269K	\$364K



PRO TIP: When entering data manually into a spreadsheet, try to stick to a tabular format – you can later tweak how you display the results in a pivot table!

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CREATING A PIVOT TABLE

Pivot Tables are created from a *single* data range with clear column headers that include both dimensions & measures:

- **Dimensions** are categorical fields typically used to *group* or *filter* data
- **Measures** are numerical fields typically used for *aggregation* or *value calculations*

Insert > PivotTable

PivotTable from table or range

Select a table or range

Table/Range: 'Sales Pipeline'!\$A\$1:\$O\$8801

Choose where you want the PivotTable to be placed

☒ New Worksheet

☐ Existing Worksheet

Location:

Choose whether you want to analyze multiple tables

☐ Add this data to the Data Model

OK Cancel

What data are you analyzing?

Where will the PivotTable live?



HEY THIS IS IMPORTANT!

Checking “**Add this data to the Data Model**” will load the data into Excel’s data model, and require the use of advanced tools like Power Pivot and DAX (covered in a separate course)

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THE FIELD LIST

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The **Field List** shows all the columns in your data

Filters appear as drop-down lists, and can be used to filter the entire pivot table view

Fields included as **Columns** will appear as individual column items in the pivot table

Fields included as **Rows** will appear as individual row items in the pivot table

Fields included as **Values** will be summarized based on Pivot Table context (*filters, rows & columns*)

The screenshot shows the 'PivotTable Fields' task pane. At the top, there's a search bar and a list of fields: Opportunity ID, Sales Agent, Sales Manager, Region, Product, Series, and Account. Below this, there are four layout areas: Filters, Columns, Rows, and Values. The Filters area contains 'Dimensions'. The Columns area contains 'Dimensions'. The Rows area contains 'Dimensions'. The Values area contains 'Measures'. Arrows point from the text annotations to these specific areas and the field list.

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PIVOTING DATA

Moving fields from rows to columns, or **pivoting**, changes the way data is displayed:

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The diagram shows the transformation of a data table through pivoting. On the left is the initial table with 'Region' and 'Product' as row headers and 'Sum of Close Value' as the data column. In the center is the 'PivotTable Fields' task pane. An arrow points from the 'Region' field in the 'Rows' area to the 'Columns' area. Another arrow points from the 'Product' field in the 'Rows' area to the 'Columns' area. On the right is the resulting pivoted table, where 'Sum of Close Value' is the row header, and 'Region' and 'Product' are column headers. The value for 'GTX Plus Basic' in the 'Central' region is highlighted as \$200,679 in both the original and pivoted tables.

Region	Product	Sum of Close Value
Central		\$3,346,293
	GTK 500	\$0
	GTX Basic	\$182,839
	GTX Plus Basic	\$276,512
	GTX Plus Pro	\$931,904
	GTX Pro	\$1,104,109
	MG Advanced	\$828,947
	MG Special	\$21,982
East		\$3,090,594
	GTK 500	\$0
	GTX Basic	\$117,331
	GTX Plus Basic	\$200,679
	GTX Plus Pro	\$822,168
	GTX Pro	\$1,285,647
	MG Advanced	\$655,914
	MG Special	\$8,855
West		\$3,568,647
	GTK 500	\$400,612
	GTX Basic	\$199,093
	GTX Plus Basic	\$228,084
	GTX Plus Pro	\$875,579
	GTX Pro	\$1,120,822
	MG Advanced	\$731,526
	MG Special	\$12,931

Sum of Close Value	Region	Central	East	West
Product				
GTK 500		\$0	\$0	\$400,612
GTX Basic		\$182,839	\$117,331	\$199,093
GTX Plus Basic		\$276,512	\$200,679	\$228,084
GTX Plus Pro		\$931,904	\$822,168	\$875,579
GTX Pro		\$1,104,109	\$1,285,647	\$1,120,822
MG Advanced		\$828,947	\$655,914	\$731,526
MG Special		\$21,982	\$8,855	\$12,931

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SUMMARIZE VALUES BY

The **summarize values by** option determines how your data will be aggregated when rolled up or summarized at different levels (*sum, count, average, max, etc.*)

RIGHT-CLICK

	Total Sales	Deals Won	Avg Deal Size	Best Sale
Month	Sum of Close Value	Count of Close Value	Average of Close Value	Max of Close Value
Mar	\$1,134,672	531	\$2,137	\$25,897
Apr	\$721,932	285	\$2,533	\$25,288
May	\$1,025,713	438	\$2,342	\$6,310
Jun	\$1,338,466	531	\$2,521	\$30,288
Jul	\$696,932	308	\$2,263	\$6,100
Aug	\$1,050,059	446	\$2,354	\$25,791
Sep	\$1,235,264	503	\$2,456	\$29,166
Oct	\$731,980	279	\$2,624	\$27,971
Nov	\$938,943	406	\$2,313	\$26,186
Dec	\$1,131,573	511	\$2,214	\$29,220
Grand Total	\$10,005,534	4,238	\$2,361	\$30,288

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SHOW VALUES AS

Show values as options allow you to apply calculations which change the way values are represented in the pivot table (*percent of column, row, total, etc.*)

RIGHT-CLICK

	No Calculation	% of Column Total	Diff. from prev. Month	Running Total
Month	Sum of Deal Won	Sum of Deal Won2	Sum of Deal Won3	Sum of Deal Won4
Mar	531	12.53%		531
Apr	285	6.72%	-246	816
May	438	10.34%	153	1,254
Jun	531	12.53%	93	1,785
Jul	308	7.27%	-223	2,093
Aug	446	10.52%	138	2,539
Sep	503	11.87%	57	3,042
Oct	279	6.58%	-224	3,321
Nov	406	9.58%	127	3,727
Dec	511	12.06%	105	4,238
Grand Total	4238	100.00%		

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SORTING

There are two ways to **sort** your pivot table data in ascending or descending order:

The diagram shows two methods to sort pivot table data. On the left, 'Items in rows or columns' shows a pivot table with 'Sales Manager' in the column header and 'Total Sales' in the row header. A dropdown menu for 'Total Sales' is open, showing options: 'Sort A to Z', 'Sort Z to A', and 'More Sort Options...'. An arrow points from this menu to a 'Sort (Sales Manager)' dialog box. On the right, 'Data in values' shows a pivot table with 'Sales Manager' in the column header and 'Total Sales' in the row header. A dropdown menu for 'Total Sales' is open, showing options: 'Sort A to Z', 'Sort Z to A', and 'More Sort Options...'. An arrow points from this menu to a 'Sort (Sales Manager)' dialog box. The dialog box has 'Sort options' with 'Manual (you can drag items to rearrange them)' selected. Below it, 'Ascending (A to Z) by:' is selected with 'Sales Manager' in the dropdown. 'Descending (Z to A) by:' is also selected with 'Total Sales' in the dropdown. The dialog box has 'OK' and 'Cancel' buttons.

Sales Manager	Total Sales
Summer Sewald	\$1,964,750
Rocco Neubert	\$1,960,545
Melvin Marxen	\$2,251,930
Dustin Brinkmann	\$1,094,363
Celia Rouche	\$1,603,897
Cara Losch	\$1,130,049

Sales Manager	Total Sales
Melvin Marxen	\$2,251,930
Summer Sewald	\$1,964,750
Rocco Neubert	\$1,960,545
Celia Rouche	\$1,603,897
Cara Losch	\$1,130,049
Dustin Brinkmann	\$1,094,363

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FILTERING

Like normal tables, you can **filter** pivot table data by values (*manual*) or by condition:

The diagram shows two methods to filter pivot table data. On the left, 'MANUAL SELECTION' shows a pivot table with 'Sales Agent' in the column header and 'Total Sales' in the row header. A dropdown menu for 'Total Sales' is open, showing options: 'Sort A to Z', 'Sort Z to A', 'More Sort Options...', 'Label Filters', and 'Value Filters'. An arrow points from 'Label Filters' to a 'Label Filter (Sales Agent)' dialog box. On the right, 'Label Filter (Sales Agent)' shows a dialog box with 'Show items for which the label' set to 'begins with' and 'M' in the input field. Below it, 'Use ? to represent any single character' and 'Use * to represent any series of characters' are shown. The dialog box has 'OK' and 'Cancel' buttons. An arrow points from the dialog box to a pivot table showing filtered data.

Sales Agent	Total Sales
Markita Hansen	\$328,792
Marty Freudenburg	\$291,195
Maureen Marciano	\$350,395
Moses Frase	\$207,182



PRO TIP: By default, you can only apply *one* filter at a time. To enable multiple filters, go to *PivotTable Analyze > Options > Totals & Filters* and select **“Allow multiple filters per field”**

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FILTERING

Like normal tables, you can **filter** pivot table data by values (*manual*) or by condition:

Label Filters:

- Equals...
- Does Not Equal...
- Begins With...
- Does Not Begin With...
- Ends With...
- Does Not End With...
- Contains...
- Does Not Contain...
- Greater Than...
- Greater Than Or Equal To...
- Less Than...
- Less Than Or Equal To...
- Between...
- Not Between...

Value Filters:

- Equals...
- Does Not Equal...
- Greater Than...
- Greater Than Or Equal To...
- Less Than...
- Less Than Or Equal To...
- Between...
- Not Between...
- Top 10...

Top 10 Filter (Sales Agent)

Show: Top 5 Items by Total Sales

Sales Agent **Total Sales**

Cassey Cress	\$450,489
Darcel Schlecht	\$1,153,214
Donn Cantrell	\$445,860
Kary Hendrixson	\$454,298
Vicki Laflamme	\$478,396



PRO TIP: By default, you can only apply *one* filter at a time. To enable multiple filters, go to *PivotTable Analyze > Options > Totals & Filters* and select **“Allow multiple filters per field”**

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PRO TIP: SLICERS & TIMELINES

Slicers are user-friendly, interactive versions of Pivot Table filters, and **Timelines** are slicers specifically designed to work with date fields

PivotTable Analyze > Insert Slicer

Insert Slicers

- ☐ Opportunity ID
- ☐ Sales Agent
- ☐ Sales Manager
- ☐ Region
- ☐ Product
- ☒ **Series**
- ☐ Account
- ☐ Sector
- ☐ Location
- ☐ Deal Stage

Series

GTK GTX MG

Product **Total Sales**

GTX Basic	\$499,263
GTX Plus Basic	\$705,275
GTX Plus Pro	\$2,629,651
GTX Pro	\$3,510,578
Grand Total	\$7,344,767



PRO TIP: Use the “Report Connections” option in the Slicer tab to link slicers to **multiple Pivot Tables** that share the same source data

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CALCULATED FIELDS

Calculated fields allow you to create measures based on existing fields

PivotTable Analyze

Fields, Items, & Sets ▾

- Calculated Field...
- Calculated Item...
- Solve Order...
- List Formulas

Insert Calculated Field

Name: Price Realization [Modify]

Formula: ="Close Value" / Sales Price [Delete]

Deal Won 1 ▾

Sales Manager ▾	Sum of Sales Price	Sum of Close Value	Price Realization
Cara Losch	\$1,130,734	\$1,130,049	99.94%
Celia Rouche	\$1,607,106	\$1,603,897	99.80%
Dustin Brinkmann	\$1,103,493	\$1,094,363	99.17%
Melvin Marxen	\$2,248,701	\$2,251,930	100.14%
Rocco Neubert	\$1,956,632	\$1,960,545	100.20%
Summer Sewald	\$1,976,816	\$1,964,750	99.39%
Grand Total	\$10,023,482	\$10,005,534	99.82%



HEY THIS IS IMPORTANT!

Calculated fields are always based on the SUM of the fields they contain